

Sofya Sellak

<https://sofyaslk.github.io/>

Education

University of Texas at El Paso *Anticipated Spring 2028*
Doctor of Philosophy in Mechanical Engineering
El Paso, TX, United States

Al Akhawayn University in Ifrane *Fall 2024*
Bachelor of General Engineering, Minor in AI
Ifrane, Morocco

University of Malaya (Exchange Program) *Spring 2022*
Faculty of Computer Science
Kuala Lumpur, Malaysia

Technical University of Ostrava (Exchange Program) *Fall 2021*
Faculty of Mechanical Engineering
Ostrava, Czech Republic

Honors and Awards

- 1st Prize in Hackathon by African Computer Vision Summer School *Summer 2024*
- Best Capstone Project of the Year (2023-2024) *Summer 2024*
- 1st Prize in AUI's Engineering Design Competition *Spring 2023*
- 1st Prize in Hack2Impact Hackathon by AUI ITSservices x Huawei *Spring 2023*
- Female Role Model in STEM by 1 Million Women in STEM *Spring 2020*
- VEX U Robotics Morocco champions *Fall 2019*

Work Experience

University of Texas at El Paso *Sep 2024 – Present*
Role: Research Assistant

AI Movement, International Artificial Intelligence Center of Morocco *Sep 2023 – Dec 2023*
Role: Research Intern

Leyton *Sep 2022 – May 2024*
Role: R&D Scientific Consultant

Digigirlz Mentorship Program (Microsoft) *Feb 2020 – Oct 2020*
Role: Mentor

AUI Mechatronics Club *Jan 2020 – Jun 2021*
Role: Software Team Member

Presentations and Conferences

Conference: IEEE International Conference on Signal, Image, Video, and Communications (ISIVC) *Spring 2024*

Title: Monocular Visual Odometry in Mobile Robot Navigation

Conference: IEEE Congress on Information Science and Technology (CiSt) *Spring 2023*

Title: A UWB Vibrotactile Safety Monitoring System for Factory Environments

Conference: IEEE International Conference on Mechatronics and Automation (ICMA) *Fall 2023*

Title: Safety Management System for Factory Workers using UWB Indoor Positioning

Publications

Journal: International Journal of Information Science and Technology *Published Sep 12, 2024*

First Author

Title: Performance Evaluation of a UWB Vibrotactile Safety Monitoring System for Factory Environments.

Skills

Robotics: Simulation, ROS, Inverse Kinematics, Kalman Filter.

Deep Learning: Vision (Mask-RCNN, etc.), VAE, Transformers, RL (PPO, etc.).

Software: C++, C, Python (PyTorch, OpenCV, SciPy, etc.), Git, Linux, VHDL.

Electrical: FPGA, ModelSim, ESP32, Arduino, NVIDIA Jetson Nano, Raspberry Pi.

Languages: Arabic (fluent), English (fluent), French (fluent).

Trainings

African Computer Vision Summer School *Jul 2024*

Covered deep learning, 3D geometry, scene understanding, and generative models.

AI Summer School *Jul 2023*

Focused on transformers, LLMs, generative AI models, and advances in robotics.